Malaviya National Institute of Technology Jaipur
Department of Computer Science and Engineering

Cryptanalysis of the Shift Cipher

Brute force with dictionary scoring and Chi-Square analysis

Course	Cryptography Laboratory (22CPP307)
Assignment	Assignment 4 (Cryptanalysis of Shift Cipher)
Group	Group 10 (Even group number)
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Instructor	Dr. Meenakshi Tripathi
Submission date	15 October 2026
Repository folder	attacks/shift_cipher_attack/

1. Purpose

The Shift (Caesar) cipher encrypts each letter as C = (P + k) mod 26 with a key k in 0..25. Its key space contains only 26 keys, so it can be broken by exhaustive search as long as the analyst can recognise the correct plaintext. This laboratory implements two ways of *recognising* English automatically - **dictionary scoring** and **Chi-Square letter-frequency analysis** - compares the keys they predict, studies when each of them fails, and proposes improvements.

2. Algorithms

2.1 Brute force with dictionary scoring

```
Input : ciphertext C, word list D (english_words.txt)
for k = 0 .. 25:
    P_k   = decrypt(C, k)                  # P = (C - k) mod 26
    words = tokenize(P_k)                  # maximal runs of letters
    score[k] = number of words that are in D
predicted_key = argmax_k score[k]          # ties broken by match percentage
```

2.2 Chi-Square frequency analysis

```
Input : ciphertext C, English letter probabilities p[A..Z]
for k = 0 .. 25:
    P_k = decrypt(C, k);  N = number of letters in P_k
    for each letter i:  O_i = count of i in P_k,   E_i = N * p[i]
    chi2[k] = sum_i (O_i - E_i)^2 / E_i
predicted_key = argmin_k chi2[k]           # smallest distance from English
```

3. Implementation

File (attacks/shift_cipher_attack/)	Role
src/shift_cipher.py	encrypt() / decrypt() with key reduction mod 26; non-letters are preserved
src/brute_force_dictionary.py	load_dictionary(), dictionary_score(), dictionary_attack() over all 26 keys
src/chi_square_attack.py	English letter probabilities, calculate_chi_square(), chi_square_attack()
src/main.py	Runs the five test cases, prints the comparison table, saves outputs/results_table.txt, interactive menu
dictionary/english_words.txt	370,105-word English list used for scoring
testcases/test_cases.json	Five test cases (normal, long, short, no spaces, rare letters)

Git history follows the required pattern: *Implemented Shift Cipher → Added brute-force and dictionary scoring → Added Chi-Square cryptanalysis →* further commits for test cases, edge cases and documentation.

4. Experimental results

Test case	Actual key	Dictionary key	Chi-Square key	Dictionary correct?	Chi-Square correct?
1. Standard Medium Text with Spaces	7	7	7	YES	YES
2. Long Technical Text	15	15	15	YES	YES
3. Short Text (3 words)	3	3	3	YES	YES
4. Continuous Text Without Spaces	12	0	12	NO	YES
5. Unusual Letter Frequencies (High Z/J/Q/X counts)	19	19	25	YES	NO

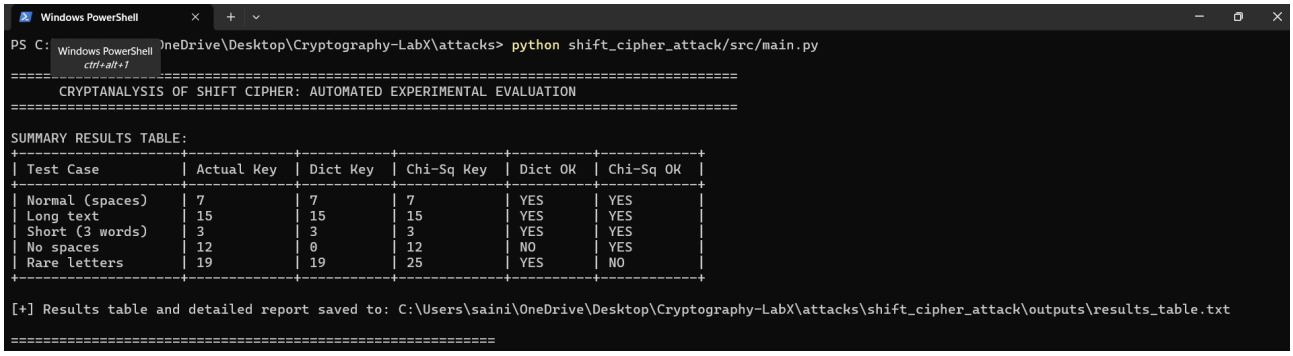


Figure 1 - Automated evaluation (python shift_cipher_attack/src/main.py)

4.1 Ciphertexts and recovered plaintexts

#	Ciphertext	Recovered plaintext (correct key)
1	AOL XBPJR IYVDU MVE QBTWZ VCLY AOL SHGF KVN HUK YBUZ HDHF ZHMLSF	THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG AND RUNS AWAY SA...
2	RGNEIPCPANHXH XH IWT HIJSN DU PCPANOXCV XCUDGBPIXDC HNHITBH XC DGSTG ID HIJSN IWT WXSSTC PHETRIH DU IWT HNHITBH PCS QGTPZ RXEWTGH	CRYPTANALYSIS IS THE STUDY OF ANALYZING INFORMATION SYSTEMS ...
3	GHIHQG WKH FDVWOH	DEFEND THE CASTLE
4	PQRQZPFTQQMEFIMXXARFTQOMEFXQMFPMIZ	DEFENDTHEEASTWALLOFTHECASTLEATDAWN
5	CTSS KARMAF UNSS JNBS OXQ	JAZZ RHYTHM BUZZ QUIZ VEX

5. Comparison of the dictionary key and the Chi-Square key

On the five test cases the two methods agree with each other and with the real key on three cases (1, 2, 3). They disagree on the two stress cases, and in each of those exactly one method is right:

- **Test 4 (no spaces):** dictionary scoring predicted key 0, Chi-Square predicted the true key 12.
- **Test 5 (JAZZ RHYTHM BUZZ QUIZ VEX):** dictionary scoring predicted the true key 19, Chi-Square predicted 25.

Property	Dictionary scoring	Chi-Square analysis
What it measures	Number of recognisable English words	Distance of letter counts from English frequencies

Property	Dictionary scoring	Chi-Square analysis
Needs word boundaries (spaces)	Yes - tokenisation depends on them	No - works on letters only
Needs a reasonable amount of text	Works on very short text if words are common	Needs roughly 40+ letters
Sensitive to unusual letters	No	Yes (J, Q, X, Z distort the statistic)
Cost	26 x (words) dictionary look-ups	26 x (letters) arithmetic

6. Failure analysis

6.1 Dictionary attack failed on Test 4 (predicted 0, actual 12)

The ciphertext `PQRQZPFTQQMEFIMXXARFTQOMEFXQMFPMIZ` has no spaces, so the tokeniser sees one 34-letter 'word'. No key makes that string a dictionary word, every key scores 0, and the tie-break returns key 0. **Improvement:** segment the candidate text into words with dynamic programming over the dictionary (maximum-match segmentation), or score with letter bigrams/trigrams instead of whole words.

6.2 Chi-Square attack failed on Test 5 (predicted 25, actual 19)

The plaintext is only 21 letters long and is packed with rare letters (J, Z, Q, X). For the true key 19 the statistic is $\chi^2 = 1726.2$, while the wrong key 25 gives $\chi^2 = 33.7$, so the wrong key looks 'more English'. **Improvement:** combine both scores (for example, rank by dictionary hits and break ties by χ^2), or use a longer sample / bigram statistics.

7. Observations from our own experiments

To see how reliability depends on length we encrypted random substrings of the repository datasets with random keys (seed 10, 200 trials per length) and attacked each ciphertext with both methods:

Ciphertext length	Trials	Dictionary correct	Chi-Square correct
10 characters	200	55.0 %	62.5 %
20 characters	200	95.0 %	85.5 %
40 characters	200	99.5 %	95.0 %
80 characters	200	100.0 %	99.5 %
160 characters	200	100.0 %	100.0 %

- Chi-Square accuracy rises quickly with length and is close to perfect from about 80 letters; very short samples are unreliable.
- Dictionary scoring is strong whenever spaces are present, even for short strings, but gives no information when they are missing.
- Both attacks finish in well under a millisecond per key, because there are only 26 keys.

8. Conclusion

The Shift cipher provides essentially no security: with 26 keys an exhaustive search is instantaneous, and either an English dictionary or a letter-frequency statistic identifies the correct key automatically. The two recognisers have complementary weaknesses - dictionary scoring needs word boundaries, Chi-Square needs enough ordinary-English letters - so a practical attack should combine them.

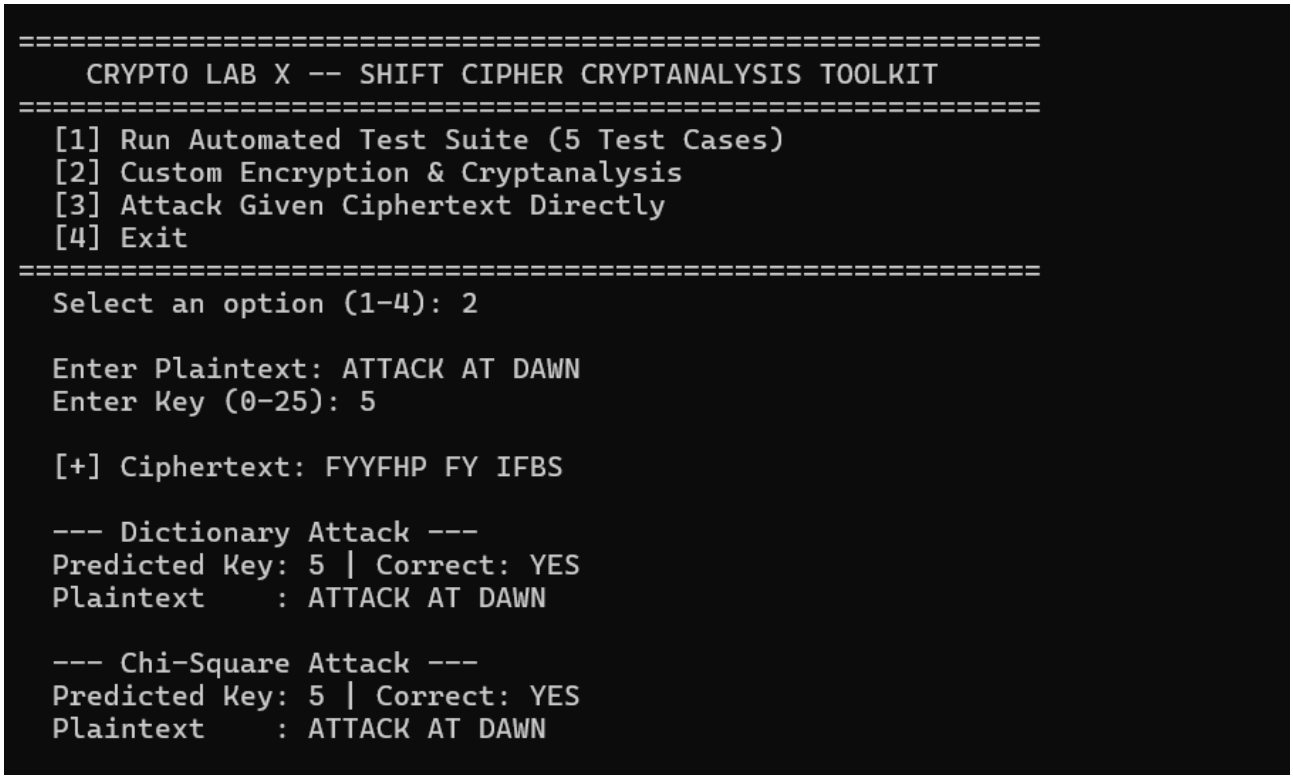


Figure 2 - Custom plaintext encrypted and attacked

```
============================================================
    CRYPTO LAB X -- SHIFT CIPHER CRYPTANALYSIS TOOLKIT
============================================================
  [1] Run Automated Test Suite (5 Test Cases)
  [2] Custom Encryption & Cryptanalysis
  [3] Attack Given Ciphertext Directly
  [4] Exit
============================================================
  Select an option (1-4): 3

  Enter Ciphertext:  FYYFHP FY IFBS

  --- Dictionary Attack Result ---
  Predicted Key: 5
  Plaintext    : ATTACK AT DAWN

  --- Chi-Square Attack Result ---
  Predicted Key: 5
  Plaintext    : ATTACK AT DAWN

============================================================
    CRYPTO LAB X -- SHIFT CIPHER CRYPTANALYSIS TOOLKIT
============================================================
  [1] Run Automated Test Suite (5 Test Cases)
  [2] Custom Encryption & Cryptanalysis
  [3] Attack Given Ciphertext Directly
  [4] Exit
============================================================
  Select an option (1-4): 4

  Exiting Cryptanalysis Toolkit. Goodbye!

PS C:\Users\saini\OneDrive\Desktop\Cryptography-LabX\attacks>
```

Figure 3 - Attacking an unknown ciphertext